JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATE, RTMNU & MSBTE Mumbai

Education to Eternity

Basic Science & Humanities Department

2023-24 (Semester)

CLASS TEST- I EXAMINATION

Course: B. Tech.

Subject: Basic Electrical & Electronics Engg.

Subject Code:

Year/Semester: I Semester (First Year)

Max Marks: 20 **Date: - 29/09/2023** **Duration:- 1 Hr**

Instructions to the Students:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

CO1.Define fundamentals of electrical system and choose measuring instruments for measurement of electrical quantities & describe the concept of PN junction diode and its characteristics.

CO2. Classify wiring system and compare energy resources for electrical energy generation & elaborate the transistor configuration in CE, CB & CC mode.

CO3. Plan and organize the utilization of energy resources of electrical system & apply transistor characteristics to construct amplifiers devices.

$R = V/I$

CO4. Compare different sources of electrical system & distinguish various logic gates and simplify the Boolean's equations.

CO5. Justify the utilization of various electrical and electronics components into electrical and electronics circuitries.

CO6. Construct various circuits using resistors, capacitors, inductors, PN junction diode, Zener diode, transformers, transistors and logic gates.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q1	Define Electric Potential.	1/CO1	1
Q2	What is specific resistivity? Give its S.I. unit.	2/CO1	1
Q3	State one ohm resistance.	1/CO2	1
Q4	Define temperature coefficient of resistance.	2/CO2	1

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SECTION B

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence)

Q.6	List different types of Electric current.	1/CO1	2
Q.7	A 100 V lamp has a hot resistance of 250Ω. Find the current taken by the lamp and its power rating in watts. $P = I^2 R$	1/CO1	2
Q.7	Write any two alloys of resistor. Also mention its important property.	2/CO2	2
Q.8	Draw the color coding diagram for 230kΩ resistor with 5% tolerance.	2/CO3	2

SECTION C

1. **Solve any two questions.**

Q.9	Derive the equation for the temperature coefficient of resistance. Mention its S.I units.	3/CO5	4
Q.10	Explain in detail the non-linear types of resistors.	3/CO2	4
Q.11	Mention different types of Electrical wiring system. And explain anyone in detail.	4/CO4	4

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